

# BHOOMI OS — AGRICULTURAL EVIDENCE LAYER

Automated Sentinel-2 & Meteorological Proof Packet for PMFBY Claims

PACKET ID: PKT-1

CLAIM DATE: 28/8/2026

STATUS: VERIFIED EVIDENCE

## 1. FARMER & AGRISTACK REGISTRY DETAILS

|               |            |                    |               |
|---------------|------------|--------------------|---------------|
| Farmer Name:  | Ravi Kumar | AgriStack ID:      | AGR-IND-88219 |
| Phone Number: | 9876543210 | Farmer ID (Local): | FRM-1         |
| Crop Sown:    | Wheat      | Total Area:        | 2.35 Hectares |

## 2. SENTINEL-2 L2A SPECTRAL AUDIT (COPERNICUS SATELLITE)

|                     |                                  |                     |                          |
|---------------------|----------------------------------|---------------------|--------------------------|
| Canopy NDVI:        | 0.22 (Vegetation Index)          | Baseline Expected:  | 0.500 NDVI               |
| Water Index (NDWI): | 0.08 (Moisture Index)            | Vegetation Deficit: | 56% Below Norm           |
| Satellite Mission:  | Copernicus Sentinel-2 Harmonized | Spatial Resolution: | 10m Multi-spectral Pixel |

## 3. METEOROLOGICAL TELEMETRY AUDIT (OPEN-METEO 7-DAY)

|                     |         |                    |                     |
|---------------------|---------|--------------------|---------------------|
| Cumulative 7d Rain: | 12.5 mm | Drought Threshold: | < 20.0 mm (Deficit) |
| Avg Max Temp:       | 37.8 °C | Heat Index Stress: | HIGH (Exceeds 35°C) |

## 4. AUTOMATED STRESS DETERMINATION & PMFBY CLAIM ADVICE

**STRESS: DROUGHT**

Triggered Rule: R1\_drought\_low\_ndvi\_dry\_spell

Assessed Crop Loss: 50% Estimated Yield Loss      Confidence Score: 88% Verified

*Evidence Summary: Sensed multi-band reflectance exhibits critical vegetative moisture collapse coinciding with unbroken dry spell and high thermal radiation during reproductive crop phase.*

## 5. CRYPTOGRAPHIC VERIFICATION & AUDIT HASH

### SHA-256 EVIDENCE INTEGRITY HASH (TAMPER-EVIDENT):

0x49882d611e2a462c3d7d6f8b6010dc259e8ce85652c676d7211f034f829e78d4